

MAΘ Competition Team HW Set 15

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Problem 1. State the definition of a rigid transformation.

Problem 2. For how many of the following types of quadrilaterals does there exist a point in the plane of a quadrilateral that is equidistant from all four vertices of the quadrilateral?

1. A square
2. A rectangle that is not a square
3. A rhombus that is not a square
4. A parallelogram that is not a rectangle or a rhombus
5. An isosceles trapezoid that is not a parallelogram

Problem 5. Isosceles $\triangle ABC$ has a right angle at C . Point P is inside $\triangle ABC$, such that $PA = 11$, $PB = 7$, and $PC = 6$. Legs $\overline{AC}$ and $\overline{BC}$ have length $s = \sqrt{a + b\sqrt{2}}$, where a and b are positive integers. What is $a + b$?

